SCSIT
IV Semester

Data Base

Exc.11

JOIN

LEFT JOIN; RIGHT JOIN

University

id_City	Name
1	Skopje
2	Ohrid
3	Bitola
4	Prilep
5	Veles
6	Tetovo
7	Gostivar
8	Kumanovo
9	Debar
10	Kichevo
11	Krushevo
12	Struga
13	Resen
14	Negotino
15	Kavadarci
16	Gevgelija
17	Strumica
18	Stip
19	Radovish
20	Kochani
21	Kratovo
22	Probistip
23	Berovo
24	Pehchevo
25	Sveti Nikole
26	Delchevo
27	Valandovo
28	Kriva Palanka
29	Makedonski Brod

id_Employee	Name	Profession	id_City
1	Mile Mileski	DBA	1
2	Maja Sazdova	Treasurer	2
3	Aleksandar Popov	Accountant	3
4	Gorjan Pehchev	Driver	3
5	Emma Petrova	Developer	4
6	Kristijan Petrovi.	Manager	1
7	Aleksandar Stefanoski	Event manager	1
8	Kosta Mickov	Manager	5
9	Ana Mircevska	Event manager	1
10	Ilina Blagoev	Guardian	10
11	Ivo Tomovski	Secretary	1
12	Ljupka Ristovska	Accountant	1

Work_place, city, vacation and employer

ID_PLace	TITLE	SALARY	SECTOR	PHONE	VEHICLE	LAPTOP
1	Manager	33500	Marketing	1	1	1
2	DBA	27000	IT	0	0	1
3	Analyst	28500	IT	1	0	1
4	Developer	25000	IT	0	0	1
5	Secretary	12500	Administration	0	0	0
6	Hygienist	7000	Administration	0	0	0
7	Accountant	10500	Finance	1	0	0
8	Treasurer	11500	Finance	1	0	0
9	Driver	7500	Service	1	1	0
10	Event manager	21500	Marketing	1	1	1
11	Guardian	14500	Administration	1	0	0
12	Associate	NULL	NULL	0	0	0
13	Waiter	1000	Catering	0	0	0

Id_Employee	FreeDays
1	10
4	15
3	22
5	21
7	5
2	7

JOIN - θ $\bowtie$

Complex product with condition

General form:

$$R\ \theta\ S = R\ \mathbf{X}_F\ S - \boldsymbol{F}\textit{-condition}$$

$$\sigma_F(R x S) - \text{JOIN through SELECTION}$$

Characteristics:

1. Commutability
2. The result contains rows from R X S that satisfy the F condition

Types of JOIN

- (Natural) Join = Inner join
- **LEFT JOIN**
- **RIGHT JOIN**

Syntax

SELECT [*, *columns*]

FROM Table1 [**LEFT, RIGHT**] **JOIN** Table2

ON Table1.Column*i* = Table2.Column*j*

[where ….]

ON = JOIN condition

Natural JOIN = SELECT

- Find the name of the town per worker

SELECT NAME_EMP, CITY_NAME

FROM EMPLOYER1, CITY

WHERE C_ID_CITY=ID_CITY

=

SELECT NAME_EMP, CITY_NAME

FROM CITY **JOIN** EMPLOYER1

ON C_ID_CITY=ID_CITY

NAME_EMP	CITY_NAME
Mile Mileski	Skopje
Maja Sazdova	Ohrid
Aleksandar Popov	Bitola
Gorjan Pehchev	Bitola
Emma Petrova	Prilep
Kristijan Petrovi.	Skopje
Aleksandar Stefanoski	Skopje
Kosta Mickov	Veles
Ana Mircevska	Skopje
Ilina Blagoev	Kichevo
Ivo Tomovski	Skopje
Ljupka Ristovska	Skopje

Where are the towns that don’t have workers?

- Find the number of workers per town (towns that don’t have workers should be on the list, but with 0 as number of workers)

SELECT CITY_NAME, COUNT(ID_EMPLOYER) AS NO_EMP

FROM CITY,EMPLOYER1

WHERE C_ID_CITY=ID_CITY

GROUP BY ID_CITY, CITY_NAME

ORDER BY NO_EMP

or

SELECT CITY_NAME, COUNT(ID_EMPLOYER) AS NO_EMP

FROM CITY **JOIN** EMPLOYER1

ON C_ID_CITY=ID_CITY

GROUP BY ID_CITY, CITY_NAME

ORDER BY NO_EMP

	CITY_NAME	NO_EMP
1	Kichevo	1
2	Ohrid	1
3	Prilep	1
4	Veles	1
5	Bitola	2
6	Skopje	6

Table1 **LEFT Join** *Table2*

- OPERATION WHICH RESULT CONTAINS **ALL** ROWS FROM TABLE1 *(LEFT TABLE)* AND ROWS (OR GROUP FUNCTIONS) FROM *TABLE2* THAT SATISFY THE JOIN CONDITION (**ON**)

Table1 ***RIGHT*** **Join** *Table2: the result is: all rows from Table2 and those rows from* Table1 that satisfy the join condition

Solution: LEFT

```
SELECT CITY_NAME,
        COUNT(ID_EMPLOYER)  AS NO_EMP
FROM CITY LEFT JOIN EMPLOYER
ON C_ID_CITY=ID_CITY
GROUP BY ID_CITY, CITY_NAME
ORDER BY  NO_EMP DESC
```

CITY_NAME	NO_EMP
Skopje	6
Bitola	2
Prilep	1
Veles	1
Ohrid	1
Kichevo	1
Krushevo	0
Struga	0
Resen	0
Negotino	0
Kavadarci	0
Gevgelija	0
Strumica	0
Stip	0
Radovish	0
Kochani	0
Kratovo	0
Probistip	0
Berovo	0
Pehchevo	0
Sveti Nikole	0
Delchevo	0
Valandovo	0
Kriva Pala...	0
Makedons...	0
Tetovo	0
Gostivar	0
Kumanovo	0
Debar	0

Solution: RIGHT

```
SELECT CITY_NAME,
        COUNT(ID_EMPLOYER)  AS NO_EMP
FROM  EMPLOYER RIGHT JOIN CITY
ON C_ID_CITY=ID_CITY
GROUP BY ID_CITY, CITY_NAME
ORDER BY  NO_EMP DESC
```

CITY_NAME	NO_EMP
Skopje	6
Bitola	2
Prilep	1
Veles	1
Ohrid	1
Kichevo	1
Krushevo	0
Struga	0
Resen	0
Negotino	0
Kavadarci	0
Gevgelija	0
Strumica	0
Stip	0
Radovish	0
Kochani	0
Kratovo	0
Probistip	0
Berovo	0
Pehchevo	0
Sveti Nikole	0
Delchevo	0
Valandovo	0
Kriva Pala...	0
Makedons...	0
Tetovo	0
Gostivar	0
Kumanovo	0
Debar	0

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- Find the number of nonworking days (holyday) per worker

```
SELECT E.NAME as NAME_EMP,
       ISNULL(FREEDAYS,0) as NO_DAYS
FROM EMPLOYER AS E
       LEFT JOIN FREEDAYS AS FR
       ON E.ID_EMPLOYEE = FR.ID_EMPLOYEE
```

	NAME_EMP	NO_DAYS
1	Mile Mileski	10
2	Maja Sazdova	7
3	Aleksandar Popov	22
4	Gorjan Pehchev	15
5	Emma Petrova	21
6	Kristijan Petrovi.	0
7	Aleksandar Stefanoski	5
8	Kosta Mickov	0
9	Ana Mircevska	0
10	Ilina Blagoev	0
11	Ivo Tomovski	0
12	Ljupka Ristovska	0

Isnull*(COLUMN, VALUE*): if in the chosen row, the column **COLUMN** has null value, in the result **VALUE** is written instead of Null

Watch what happens if you accidentally change places of tables !!!

```
SELECT  E.NAME as NAME_EMP,
        ISNULL(FREEDAYS,0) as NO_DAYS
FROM    FREEDAYS AS FR
        LEFT JOIN
        EMPLOYER AS E
            ON E.ID_EMPLOYEE = FR.ID_EMPLOYEE
```

NAME_EMP	NO_DAYS
Mile Mileski	10
Gorjan Pehchev	15
Aleksandar Popov	22
Emma Petrova	21
Aleksandar Stefanoski	5
Maja Sazdova	7

THE SQL QUERY ***WORKS****, BUT* ***THIS IS NOT THE CORRECT ANSWER !!!***

IT DOESN'T CONTAIN ALL DATA WE NEED

- Find the number of workers per sector.

- Find and sort number of workers per working place

- Find total number of nonworking days per working place

- *Find total number of nonworking days per sector

- FIND SALLARIES PER CITY
- FIND NUMBER OF AGENTS PER CITY (*table AGENT)*